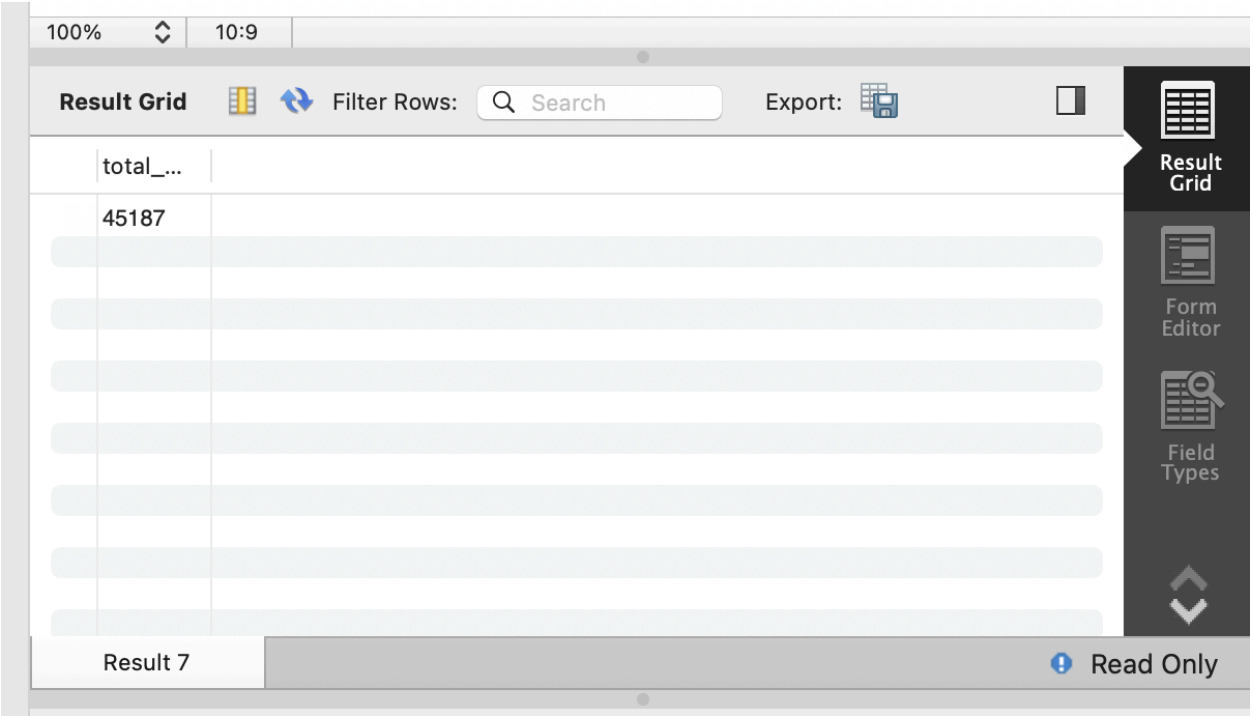
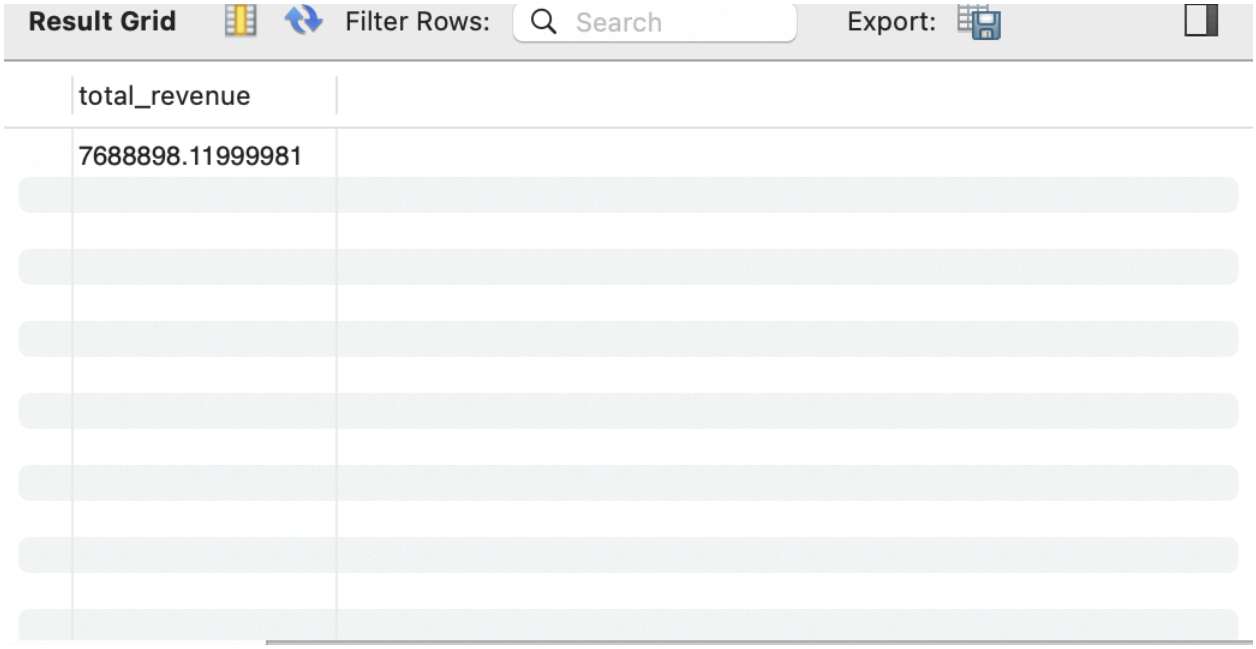
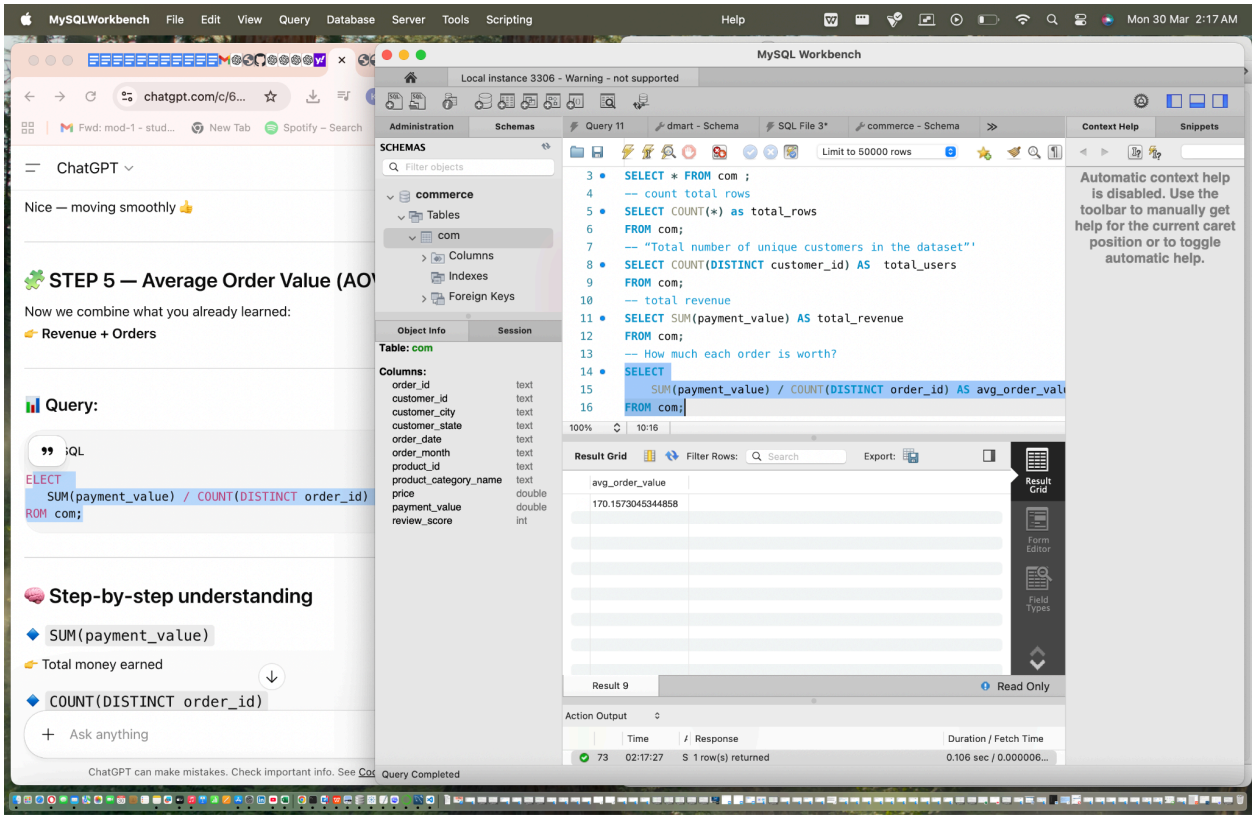
1.



2.-- total revenue
SELECT SUM(payment_value) AS total_revenue
FROM com;



3.



"Customers spend around 170 per order on average."

4 -- "How many users are active each day?

```
SELECT
order_date,
COUNT(DISTINCT customer_id) AS dau
FROM com
GROUP BY order_date
ORDER BY order_date;
```

-- "There is a significant spike in DAU on 24th November 2017, likely due to a promotional event such as Black Friday."

Result Grid Filter Rows: Search Export:

order_date	dau
2018-01-28	73
2018-01-29	106
2018-01-30	119
2018-01-31	117
2018-02-01	107
2018-02-02	107
2018-02-03	90
2018-02-04	89
2018-02-05	123
2018-02-06	125
2018-02-07	119
2018-02-08	96
2018-02-09	103
2018-02-10	84

Result 10

```
-- Product growth over time
SELECT
   order_month,
   COUNT(DISTINCT customer_id) AS mau
FROM com
GROUP BY order_month
ORDER BY order_month;
-- "MAU shows steady growth with a peak during late 2017, likely driven by seasonal promotions and increased user adoption."
```

order_month	mau
2016-09	2
2016-10	148
2016-12	1
2017-01	386
2017-02	845
2017-03	1238
2017-04	1078
2017-05	1682
2017-06	1423
2017-07	1811
2017-08	1902
2017-09	1957
2017-10	2170
2017-11	3451

```
-- DAU / MAU Ratio for User engagement
SELECT
```

```
    d.order_date,
    d.dau,
    m.mau,
    d.dau / m.mau AS engagement_ratio
FROM
(
    SELECT
        order_date,
        COUNT(DISTINCT customer_id) AS dau
    FROM com
    GROUP BY order_date
) d
JOIN
(
    SELECT
        order_month,
        COUNT(DISTINCT customer_id) AS mau
    FROM com
    GROUP BY order_month
) m
ON LEFT(d.order_date, 7) = m.order_month;
-- “The DAU/MAU ratio stabilizes between 5–10%, indicating moderate user engagement, with users returning occasionally but not daily.”
```

order_date	dau	mau	engagement_ra...
2016-09-05	1	2	0.5000
2016-09-04	1	2	0.5000
2016-10-10	19	148	0.1284
2016-10-09	17	148	0.1149
2016-10-08	23	148	0.1554
2016-10-07	17	148	0.1149
2016-10-06	22	148	0.1486
2016-10-05	20	148	0.1351
2016-10-04	27	148	0.1824
2016-10-03	3	148	0.0203
2016-12-23	1	1	1.0000
2017-01-31	34	386	0.0881
2017-01-30	27	386	0.0699
2017-01-29	21	386	0.0544

```
-- RETENTION ANALYSIS
-- First Purchase Per User
SELECT
    customer_id,
    MIN(order_date) AS first_purchase_date
```

FROM com
GROUP BY customer_id;

Result Grid Filter Rows: Search Export:

customer_id	first_purchase_d...
373373ec4d694242d601fe8a69be1c92	2017-10-05
1f68e24da06f36cb8a5243e23c390bcf	2018-05-07
9f582ffae8aecd135ae3d11fcd508c26	2017-09-20
1c3f1e66dc3859cce89cb9c12e3c1122	2017-07-21
32755045a3b602adc3876bbfa3f8d7c8	2018-04-21
10c2ef0e5a6a9b2d28d70b65b5063bf3	2018-01-19
4ca39a384a51c378303737fa14db7ae9	2018-05-09
85c6a3112d9b45e4bbb16f9337537dcd	2017-05-29
54af9d31eb26ac435fa7bc914d9a30bf	2017-10-10
6dd52a73b6f4356624e6622b1a1e1b5f	2018-04-26
4f16e4f7467c532048801d636774945f	2017-04-19
181512b829112330394f0fc0985980fe	2017-06-06
e9446b50eed1a1ef77178a47055332f1	2017-10-09

-- “I identified each user’s first purchase date to establish the starting point for retention and cohort analysis.”

5-- “How many users came back again?”

```
SELECT
   customer_id,
   COUNT(order_id) AS total_orders
FROM com
GROUP BY customer_id
HAVING COUNT(order_id) > 1;
```

-- The dataset shows very low repeat user behavior, with only around 3–4% of users making more than one purchase.

customer_id	total_orders
2326c5f3dfff28efaadce0c7e62b5a73	2
d19766c98ec819e36dc3d8db0491f72f	2
02e5db654b2f6b734ed89d5a28cb200e	2
6eeb0d55dcc3c0d2973619f1362da7f5	2
5c896117244c11cdcc73221662757e05	2
9b4daa1db65ef1623983286611eb63e1	2
96d93b5cda637617dcceb8e7059b19c9	2
3a5a9c83ac1027afafe0598a12c7d6d3	2
c6178bc9698c804aa6325efaabfd1b79	2
599db69c933e5f5a111f44e413ecff87	2
3ca12a51d893d20d10c81d3ed566c391	2
c64e6cd99ad2fafbb731fb38d4f17ca6	2
68930e4bd02e1b1e603d462c6e4e2849	2
3333f866a124d5587c7b3233bb9173d0	2

Result 24

n Output

	Time		Response	Duration
90	02:59:22	S	1543 row(s) returned	

Sign in to you

6 -- Retention Rate

```
SELECT
   COUNT(DISTINCT CASE WHEN orders > 1 THEN customer_id END)
   / COUNT(DISTINCT customer_id) AS retention_rate
FROM (
   SELECT customer_id, COUNT(order_id) AS orders
   FROM com
   GROUP BY customer_id
) t;
```

-- “The retention rate is extremely low at around 3.4%, indicating that the majority of users do not return after their first purchase.”

retent...
0.0341

```
-- Where are users dropping off?
SELECT
    COUNT(DISTINCT customer_id) AS users,
    COUNT(DISTINCT order_id) AS orders,
    SUM(payment_value) AS revenue
FROM com;
-- “There is no drop-off between users and orders, indicating that nearly every user completes a purchase, but fails to return.”
```

users	orders	revenue
45187	45187	7688898.11999981

```
--  Conversion Rates
SELECT
    COUNT(DISTINCT customer_id) AS users,
    COUNT(DISTINCT order_id) AS orders,
    SUM(payment_value) AS revenue,
```

```
    COUNT(DISTINCT order_id) * 1.0 / COUNT(DISTINCT customer_id) AS
user_to_order_conversion,

    SUM(payment_value) / COUNT(DISTINCT order_id) AS avg_revenue_per_order

FROM com;
-- “The business generates consistent revenue per order, but growth is limited due to poor
repeat purchase behavior.”
```

users	orders	revenue	user_to_order_conversi...	avg_revenue_per_or...
45187	45187	7688898.11999981	1.00000	170.1573045344858

```
7. -- Which products drive revenue?
SELECT
    product_category_name,
    COUNT(DISTINCT order_id) AS total_orders,
    SUM(payment_value) AS revenue,
    AVG(payment_value) AS avg_order_value
FROM com
GROUP BY product_category_name
ORDER BY revenue DESC;
-- computers → AOV ~1296 🔥
-- home_appliances_2 → ~514
-- fixed_telephony → ~418
-- The business relies on a combination of high-volume categories for consistent revenue and
high-value categories for profitability, while facing challenges in retention and category data
quality.”
```

product_category_na...	total_orders	revenue	avg_order_value
health_beauty	3996	669354.3000000011	163.49640937958014
bed_bath_table	4402	660844.6600000001	139.41870464135025
watches_gifts	2437	574281.4400000004	229.43725129844202
computers_accessories	3133	566553.3499999989	175.5665788658193
sports_leisure	3580	550120.7600000021	150.26516252390115
furniture_decor	3141	497345.1700000016	149.17371625674915
housewares	2765	390795.5099999997	136.16568292682916
auto	1755	322250.67999999935	179.2272969966626
cool_stuff	1578	311819.5300000007	191.6530608481873
garden_tools	1718	306470.5100000007	170.7356601671313
toys	1722	260199.33999999962	147.25486134691545
perfumery	1445	210121.4	140.83203753351205
baby	1234	207326.48000000016	163.24919685039382
office_furniture	604	199682.7999999999	314.46110236220454

8.-- TOP 5 CATEGORIES

```
SELECT
   product_category_name,
   SUM(payment_value) AS revenue
FROM com
GROUP BY product_category_name
ORDER BY revenue DESC
LIMIT 5;
```

Result Grid Filter Rows: Search Export:

product_category_na...	revenue	category_rank	revenue_share
health_beauty	669354.3000000009	1	0.08705464548410488
bed_bath_table	660844.6600000015	2	0.08594790172613202
watches_gifts	574281.4400000009	3	0.0746896929881548
computers_accessories	566553.3499999993	4	0.0736845957844476
sports_leisure	550120.7600000009	5	0.07154741178961033

> “The top 5 product categories contribute around 40% of total revenue, indicating a strong concentration in a few key segments.”

9. -- Repeat Purchase Funnel

```
SELECT
```

```
    COUNT(DISTINCT customer_id) AS total_users,

    COUNT(DISTINCT CASE WHEN order_count >= 1 THEN customer_id END) AS
first_purchase,

    COUNT(DISTINCT CASE WHEN order_count >= 2 THEN customer_id END) AS
repeat_users,

    COUNT(DISTINCT CASE WHEN order_count >= 3 THEN customer_id END) AS loyal_users

FROM (
    SELECT
        customer_id,
        COUNT(order_id) AS order_count
    FROM com
    GROUP BY customer_id
) t;
WITH user_orders AS (
    SELECT
        customer_id,
        COUNT(order_id) AS order_count
    FROM com
    GROUP BY customer_id
)

SELECT
    COUNT(*) AS total_users,

    COUNT(CASE WHEN order_count >= 2 THEN 1 END) AS repeat_users,
    COUNT(CASE WHEN order_count >= 3 THEN 1 END) AS loyal_users,

    COUNT(CASE WHEN order_count >= 2 THEN 1 END) * 1.0 / COUNT(*) AS repeat_rate,
    COUNT(CASE WHEN order_count >= 3 THEN 1 END) * 1.0 / COUNT(*) AS loyalty_rate

FROM user_orders;
-- “The repeat rate is only 3.4%, and the loyalty rate is below 1%, indicating a severe retention
problem where users are not forming long-term engagement with the product.”
```

total_...	repeat_users	loyal_users	repeat_rate	loyalty_rate
45187	1543	197	0.03415	0.00436

```
-- FIND WHERE USERS DROP user segmentation and funnel analysis
WITH user_orders AS (
   SELECT
      customer_id,
      COUNT(order_id) AS order_count
   FROM com
   GROUP BY customer_id
)

SELECT
   COUNT(*) AS total_users,

   SUM(order_count >= 1) AS step_1_users,
   SUM(order_count >= 2) AS step_2_users,
   SUM(order_count >= 3) AS step_3_users,

   SUM(order_count >= 2) * 1.0 / SUM(order_count >= 1) AS step1_to_step2,

   SUM(order_count >= 3) * 1.0 / SUM(order_count >= 2) AS step2_to_step3

FROM user_orders;
WITH user_orders AS (
   SELECT
      customer_id,
      COUNT(order_id) AS order_count,
      SUM(payment_value) AS total_spent
   FROM com
   GROUP BY customer_id
),

segmented AS (
   SELECT *,
      CASE
         WHEN order_count = 1 THEN 'One-time'
         WHEN order_count = 2 THEN 'Repeat'
         WHEN order_count >= 3 THEN 'Loyal'
```

```
        END AS user_type
    FROM user_orders
)

SELECT
    user_type,
    COUNT(*) AS users,
    AVG(total_spent) AS avg_spent,
    SUM(total_spent) AS total_revenue,
    SUM(total_spent) * 1.0 / SUM(SUM(total_spent)) OVER () AS revenue_share
FROM segmented
GROUP BY user_type;
-- “The business relies heavily on one-time users (92% of revenue), while repeat and loyal users are very limited. However, repeat and loyal users have significantly higher average spend, indicating strong potential for revenue growth through retention strategies
```

user_type	users	avg_spent	total_revenue	revenue_share
One-time	43644	162.913081523229	7110178.529999807	0.924733091664373
Repeat	1346	344.6165081723628	463853.8200000003	0.06032773653138377
Loyal	197	583.0749746192892	114865.76999999996	0.014939171804243239

cohort analysis

```
-- REVENUE COHORT
WITH first_purchase AS (
    SELECT
        customer_id,
        MIN(STR_TO_DATE(order_date, '%Y-%m-%d')) AS first_order_date
    FROM com
    GROUP BY customer_id
),

cohort_data AS (
    SELECT
        c.customer_id,
        DATE_FORMAT(f.first_order_date, '%Y-%m') AS cohort_month,
        c.payment_value
    FROM com c
    JOIN first_purchase f
        ON c.customer_id = f.customer_id
```

```
),

cohort_size AS (
    SELECT
        cohort_month,
        COUNT(DISTINCT customer_id) AS cohort_users
    FROM cohort_data
    GROUP BY cohort_month
)

SELECT
    c.cohort_month,

    COUNT(DISTINCT c.customer_id) AS users,

    SUM(c.payment_value) AS total_revenue,

    ROUND(
        SUM(c.payment_value) * 1.0 / COUNT(DISTINCT c.customer_id),
        2
    ) AS revenue_per_user   -- LTV

FROM cohort_data c
GROUP BY c.cohort_month
ORDER BY c.cohort_month;
```

cohort_mon...	users	total_revenue	revenue_per_user
2016-09	2	347.52	173.76
2016-10	148	27845.809999999987	188.15
2016-12	1	19.62	19.62
2017-01	386	71910.26999999993	186.3
2017-02	845	150349.2399999999	177.93
2017-03	1238	206248.5699999997	166.6
2017-04	1078	191802.09000000005	177.92
2017-05	1682	267077.0400000003	158.79
2017-06	1423	238917.76999999993	167.9
2017-07	1811	288082.3599999997	159.07
2017-08	1902	308050.01	161.96
2017-09	1957	362772.70000000135	185.37
2017-10	2170	393858.7800000019	181.5

com 69 | Result 70 | Result 71 | Re

```
-- North star metric
-- Users don't come back after first purchase?
WITH user_orders AS (
    SELECT
        customer_id,
        COUNT(order_id) AS order_count
    FROM com
    GROUP BY customer_id
)

SELECT
    COUNT(*) AS total_users,
    SUM(order_count >= 2) AS repeat_users,

    ROUND(
        SUM(order_count >= 2) * 1.0 / COUNT(*),
        4
    ) AS repeat_purchase_rate

FROM user_orders;
```

-- “I chose repeat purchase rate as the North Star Metric because it directly reflects user retention and long-term value. My analysis showed that only ~3.4% of users make a second purchase, indicating a major drop-off after the first transaction. Improving this metric would significantly increase lifetime value and overall revenue.”

	total_...	repeat_users	repeat_purchase_r...
	45187	1543	0.0342

]-- “Is the business overly dependent on a small group of customers?”

```
WITH user_revenue AS (
   SELECT
      customer_id,
      SUM(payment_value) AS total_spent
   FROM com
   GROUP BY customer_id
),

ranked AS (
   SELECT
      customer_id,
      total_spent,
      NTILE(10) OVER (ORDER BY total_spent DESC) AS decile
   FROM user_revenue
),

final AS (
   SELECT
      decile,
      COUNT(*) AS users,
      SUM(total_spent) AS revenue,
      SUM(total_spent) * 1.0 / SUM(SUM(total_spent)) OVER () AS revenue_share
   FROM ranked
   GROUP BY decile
)

SELECT *
FROM final
ORDER BY decile;
```

-- “I analyzed revenue concentration using decile segmentation and found that a small percentage of users contributes a disproportionately large share of revenue. This indicates

dependency risk. I would recommend focusing on retaining high-value users while also improving conversion of mid-tier users to reduce reliance on a small segment."

Result Grid Filter Rows: Search Export:

decile	users	revenue	revenue_share
1	4519	3061422.5899999985	0.39816141951949796
2	4519	1179107.9899999956	0.1533520111201572
3	4519	839415.6600000007	0.10917242586639982
4	4519	666913.7699999986	0.08673723589408137
5	4519	539599.7300000013	0.07017907138038673
6	4519	437434.62000000005	0.05689171753520388
7	4519	349694.72000000364	0.04548047256477417
8	4518	277781.08000000054	0.036127553735879205
9	4518	208339.67999999973	0.02709616862500446
10	4518	129188.27999999885	0.016801923758615087

```
–  Retention Curve
-- How many users return over time after their first purchase?
WITH first_purchase AS (
   SELECT
      customer_id,
      MIN(order_date) AS first_date
   FROM com
   GROUP BY customer_id
),

retention_data AS (
   SELECT
      c.customer_id,
      f.first_date,
      c.order_date,
      TIMESTAMPDIFF(MONTH, f.first_date, c.order_date) AS cohort_index
   FROM com c
   JOIN first_purchase f
      ON c.customer_id = f.customer_id
)

SELECT
   cohort_index,
```

```
    COUNT(DISTINCT customer_id) AS users
FROM retention_data
GROUP BY cohort_index
ORDER BY cohort_index;
```

The retention curve drops sharply after Month 0, with a very small percentage of users returning in subsequent months. This confirms that most users make only one purchase and do not come back.

cohort_ind...	users
0	45187

]

```
-- How does retention vary across different user cohorts over time
-- Do newer users behave better or worse?
-- How retention changes month by month
WITH first_purchase AS (
    SELECT
        customer_id,
        MIN(order_date) AS first_date
    FROM com
    GROUP BY customer_id
),

cohort_data AS (
    SELECT
        c.customer_id,
        DATE_FORMAT(f.first_date, '%Y-%m') AS cohort_month,
        TIMESTAMPDIFF(MONTH, f.first_date, c.order_date) AS cohort_index
    FROM com c
    JOIN first_purchase f
        ON c.customer_id = f.customer_id
)

SELECT
```

```
    cohort_month,
    cohort_index,
    COUNT(DISTINCT customer_id) AS users
FROM cohort_data
GROUP BY cohort_month, cohort_index
ORDER BY cohort_month, cohort_index;
-- Even newer cohorts do not show improved retention, which suggests the issue is not due to
specific campaigns or time periods, but rather a fundamental problem in user experience or
engagement strategy."
```

cohort_mon...	cohort_ind...	users
2016-09	0	2
2016-10	0	148
2016-12	0	1
2017-01	0	386
2017-02	0	845
2017-03	0	1238
2017-04	0	1078
2017-05	0	1682
2017-06	0	1423
2017-07	0	1811
2017-08	0	1902
2017-09	0	1957
2017-10	0	2170
2017-11	0	3451

Result 116

The business has strong user acquisition and successful first purchase conversion
Retention is extremely low with around 96 percent of users not returning
Cohort analysis shows users only appear in their first month which confirms no repeat behavior
Most revenue comes from one time users making the business dependent on constant acquisition
The core issue is not conversion but post purchase retention failure

Root causes

Users are buying products that are not frequently needed such as electronics or appliances
There is no retention strategy like discounts or loyalty programs
There is no engagement after purchase such as emails or recommendations

The post purchase experience may be weak including delivery or support issues
Users have no incentive or habit to return to the platform

Solutions

Introduce a second purchase discount within a short time after first order
Implement email and notification campaigns to re engage users
Build a loyalty program with rewards points or cashback
Improve delivery experience and customer support to build trust
Use personalized product recommendations to encourage repeat purchases
Target high value users with exclusive offers to increase lifetime value

Expected impact

Increase repeat purchase rate from around 3 percent to higher levels
Improve customer retention and reduce churn
Increase revenue from repeat users instead of relying on new users
Build long term customer relationships and sustainable growth